

Comparative Analysis

F1 Score by dataset (Accending Order)

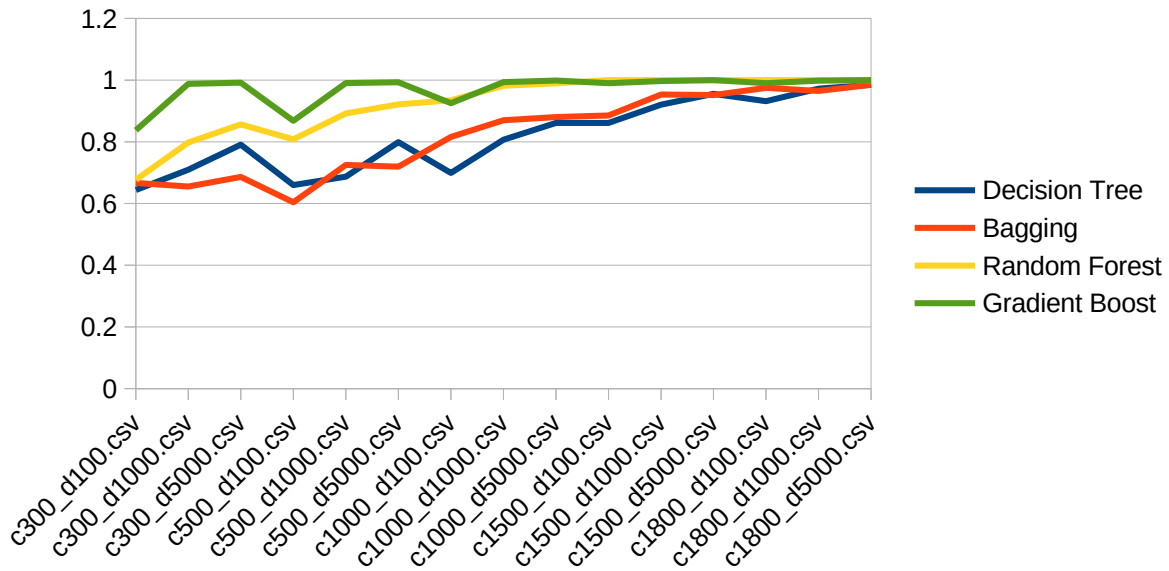


Figure 1 (F1 Scores)

Accuracy by dataset (Accending Order)

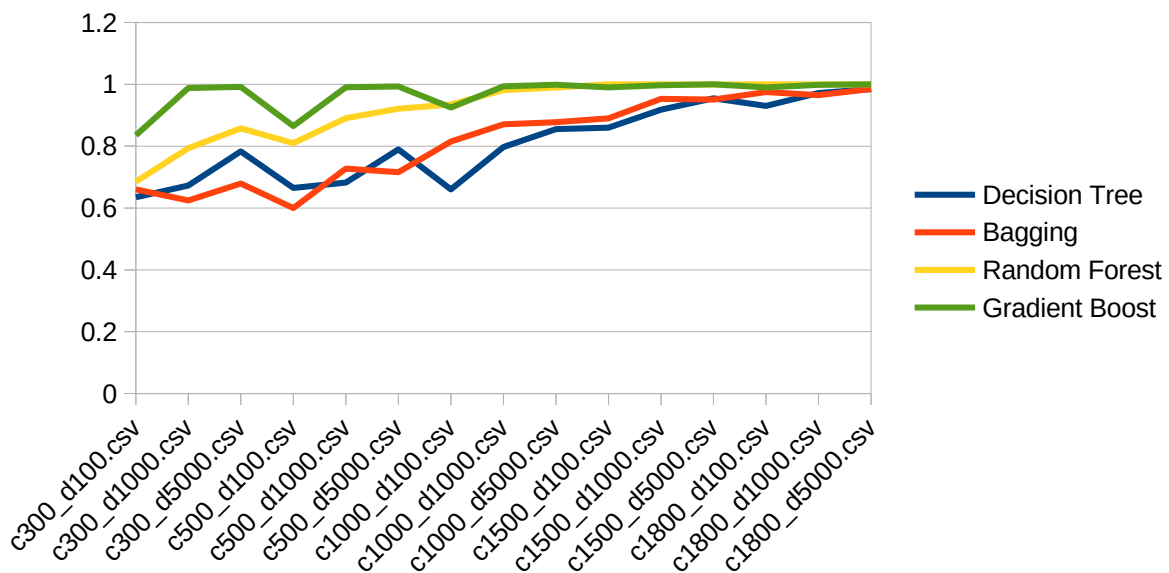


Figure 2 (Accuracy)

1. Overall many of the classifiers converged to a max f1 score of around one on the larger datasets. However, out of all of the classifiers both random forest and gradient boost stood out from the rest for their performance on smaller datasets. Looking only at these two we can see that while boosting performed better in the beginning it struggled to reach one like the random forest classifier did. Still despite this I would give the best overall performance to the boosting classifier. This is because of the good performance at lower data counts in addition to larger ones. While it didn't converge to 1 100% of the time it was often only a fraction of a point of from 1 which is still more than acceptable for the boost that it provides for the smaller datasets.
2. This question can be split into two groupings, decision trees and bagging, and random forest and gradient boost. The first group increased steadily overtime as the data size increased. This makes sense as neither of these classifiers only use the data directly and do not add anything like randomization to potentially improve the outcomes. Therefore the linear increase in accuracy and f1 makes sense. For the second group we see that they quickly converge to 1 for both accuracy and f1 at a much lower dataset then the first group. This is because we are adding additional random factors into the consideration of the data rather than just considering the data itself. This is why we see better performance out of the second group of classifiers. The second group was not effected by larger data size after a certain point.
3. Similarly to the above question there is a trend with the two groups that the more features that were apart of the dataset the more accurate and higher f1 score they achieved overall.

classifier	accuracy
DecisionTree	0.8719
Bagging	0.5853
RandomForest	0.9657
Boosting	0.9459

Figure 3 (Accuracy scores on Mnist)

1. The random forest outperformed all of the other classifiers on the minst data set. This result makes sense as boosting may reduce bias but may increase the variance resulting in an overall worse performance on the test set making sense that this might become an issue on higher sample datasets. Meanwhile Random forest has the lowest bias and variance of the classification methods used in this assignment so it makes sense that as the dataset grows so will the accuracy overall as the bias and variance are the lowest.